A MicroProject Report

On

FDP / STTP Digital Certificate Management System

Submitted to Savitribai Phule Pune University, Pune, as a part of the academic course work requirements for the completion of

Second Year Bachelor of Technology

in

Information Technology

Submitted by

Syed Afzaal	**C36**
Tanmay Walthare	**C37**
Tejas Salunke	**C38**
Viraj Thakur	**C49**

Department of Information Technology

G H Raisoni College of Engineering and Management, Pune

Academic Year: 2025-26

Summer 2026

GHraisoni COLLEGE | Engineering and Management Pune

An Empowered Autonomous Institute, Affiliated to SPPU, Pune
NAAC Accredited "A+" Grade

CERTIFICATE

This is to certify that the Microproject report entitled

FDP / STTP Digital Certificate Management System

Submitted by

Syed Afzaal	**C36**
Tanmay Walthare	**C37**
Tejas Salunke	**C38**
Viraj Thakur	**C49**

are Bonafide students of this institute and have carried out the course work of JavaScript and PHP Microproject under the supervision of **Mrs. Priyanka Kumbhar**. This report is approved for partial fulfillment of the requirement of the S.Y B.Tech. (Information Technology) for the academic year 2025–26.

Mrs. Priyanka Kumbhar
Guide

Dr. Poonam Gupta
Head of Department

Dr. N. B. Hulle
I/C Director
GHRCEM, Pune

External Examiner

Date:

Place: Pune

ACKNOWLEDGMENT

We would like to express our sincere thanks to **Dr. Poonam Gupta** Head of Information Technology Department for continuous support & valuable suggestions. We would like to thank **Mrs. Priyanka Kumbhar**, for her/his valuable support in providing us with the required information, guidance and light of knowledge, we could complete this Microproject.

We take this opportunity to thank all the staff members of Department of Information Technology Engineering for their help whenever required.

We take this opportunity to thank our Incharge Director, **Dr. N. B. Hulle** and Campus Director, **Dr. R. D. Kharadkar**, for his whole hearted support, motivation & valuable suggestions.

Finally, we would like to give special thanks to all staff members of **Department of Information Technology**, G H Raisoni College of Engineering and Management, Pune & our friends for their kind support.

Syed Afzaal	**C36**
Tanmay Walthare	**C37**
Tejas Salunke	**C38**
Viraj Thakur	**C49**

ABSTRACT

The FDP/STTP Digital Certificate Management System is a web-based application designed to manage and verify academic certificates efficiently. In traditional systems, certificate handling is done manually, which is time-consuming and prone to errors or data loss. This project provides a digital platform where users can register, log in, and upload their FDP/STTP certificates along with necessary details such as academic year, program type, and contact number.

The system stores all information securely in a database and allows administrators to review submitted certificates. The admin can approve or reject certificates, and in case of rejection, a reason is provided so the user can correct and resubmit the data. This process ensures transparency and accuracy in certificate management.

The project is developed using HTML, CSS, and JavaScript for the frontend, PHP for backend processing, and MySQL for database management. Overall, the system reduces manual effort, saves time, prevents data loss, and provides a reliable and user-friendly solution for managing academic certificates digitally.

Index

Sr. No.	Title	Page No.
1	**Introduction**	1
1.1	Overview	1
1.2	Motivation	1
1.3	Scope and Limitations	1
2	**Objective of the Project**	2
2.1	Problem Statement	2
2.2	Proposed Solution	2
3	**Tools / Technologies Used**	3-4
3.1	Frontend Technologies	3
3.2	Backend Technology	3
3.3	Database	3
3.4	Development Tools	3
3.5	Other Tools	4
3.6	GitHub link	4
4	**Project Design / Methodology**	5-9
4.1	System Architecture	5
4.2	Modules of the System	5-6
4.3	Data Flow Diagram (DFD)	6-7
4.4	Database Design	7
4.5	Tables Used	7-9
5	**Implementation / Working**	10-12
5.1	Frontend Implementation	10
5.2	Backend Implementation	10

5.3	Database Connectivity	11
5.4	Flowchart	11
5.5	Working Flow	12
6	**Results / Output**	13-17
7	**Conclusion and Future Work**	18
7.1	Conclusion	18
7.2	Future Work	18
8	**References**	19-20

1. Introduction

1.1 Overview

The FDP/STTP Certificate Management System is designed to manage academic certificates digitally. In traditional systems, certificates are handled manually, which can lead to data loss, duplication, and inefficiency. This project provides a web-based platform where users can upload certificates and administrators can manage and verify them efficiently.

1.2 Motivation

Manual certificate management is time-consuming, error-prone, and difficult to maintain. There is a need for a digital system that automates the process, ensures data security, and reduces paperwork.

1.3 Scope and Limitations

Scope:

- Online certificate submission
- Admin verification and approval
- Secure storage of certificates

Limitations:

- No mobile application
- Requires internet connection.

2. Objective of the Project

2.1 Problem Statement

In many educational institutions, the management of FDP (Faculty Development Program) and STTP (Short Term Training Program) certificates are still done manually. This traditional method involves maintaining physical records or basic digital files, which leads to several issues such as data redundancy, loss of important documents, and difficulty in tracking certificate status. Manual verification is time-consuming and prone to human errors, making the process inefficient and unreliable. Additionally, there is no proper system for users to know whether their certificates are approved or rejected, and no structured way to provide feedback or corrections.

2.2 Proposed Solution

To overcome the limitations of the manual system, a web-based FDP/STTP Digital Certificate Management System is proposed. This system allows users to register, log in, and upload their certificates along with necessary details such as program type, academic year, and contact number. The certificates are stored securely in a database with a “Pending” status. Administrators can log in to the system, review uploaded certificates, and either approve or reject them. In case of rejection, the admin provides a reason, allowing users to correct and resubmit their certificates. This automated system reduces manual work, improves accuracy, enhances transparency, and ensures efficient certificate management.

3. Tools/Technologies Used

3.1 Frontend Technologies

The frontend of the system is developed using **HTML, CSS, and JavaScript**. HTML is used to create the structure of web pages such as forms, tables, and layouts. CSS is used to design and style the user interface, making it visually appealing and user-friendly. JavaScript is used to add interactivity and performance
client-side validations such as checking empty fields and validating input data before submission.

3.2 Backend Technology

The backend of the system is developed using **PHP (Hypertext Preprocessor)**. PHP is used to handle form data, manage user sessions, and perform server-side operations such as inserting, updating, and retrieving data from the database. It also controls the application logic, such as login authentication, certificate upload processing, and approval or rejection by the admin.

3.3 Database

The system uses **MySQL** as the database management system. MySQL is used to store all application data, including user details, admin credentials, and certificate information. It provides a structured and secure way to manage data using tables and queries.

3.4 Development Tools

The project is developed using tools such as:

- **Visual Studio Code (VS Code)** for writing and editing code.
- **XAMPP Control Panel** for running Apache and MySQL services.
- **phpMyAdmin** for managing the database.

These tools help in efficient development, debugging, and testing of the project.

3.5 Other Tools

The project uses **GitHub** for version control and code management. GitHub allows storing the project code online, tracking changes, and maintaining different versions of the project. It also helps in collaboration and backup of the project files.

3.6 GitHub link:

https://github.com/VirajThakur1226/fdp-sttp-digital-certificate-management-.git

4. Project Design /Methodology

4.1 System Architecture

The FDP/STTP Digital Certificate Management System follows a **three-tier architecture** consisting of:

- **Presentation Layer (Frontend):**
  This layer is developed using HTML, CSS, and JavaScript. It provides the user interface for registration, login, certificate upload, and viewing status.
- **Application Layer (Backend):**
  This layer is implemented using PHP. It handles business logic such as authentication, certificate processing, approval/rejection, and communication between frontend and database.
- **Data Layer (Database):**
  MySQL database is used to store user information, admin details, and certificate records securely.

4.2 Modules of the System

The system is divided into two main modules:

1. User Module

- User Registration
- User Login
- Upload Certificate
- View Certificate Status (Approved / Rejected / Pending)
- Edit Certificate (if rejected)
- Download Certificate

2. Admin Module

- Admin Login
- View All Certificates
- Approve Certificates

- Reject Certificates with Reason
- Manage Users and Records

4.3 Data Flow Diagram (DFD)

Level 0 (Context Diagram)

- User interacts with the system by submitting certificate data.
- System processes data and stores it in the database.
- Admin accesses the system to verify and update certificate status.

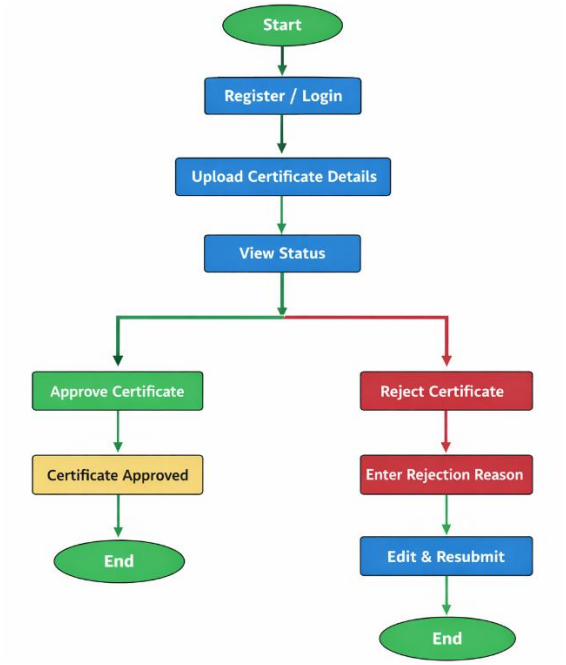


Fig. DFD Level 0 Diagram

Level 1 DFD

1. User submits certificate → Data stored in database.
2. Admin retrieves certificate → Reviews data.
3. Admin updates status (Approve/Reject)
4. Updated status is displayed to user.

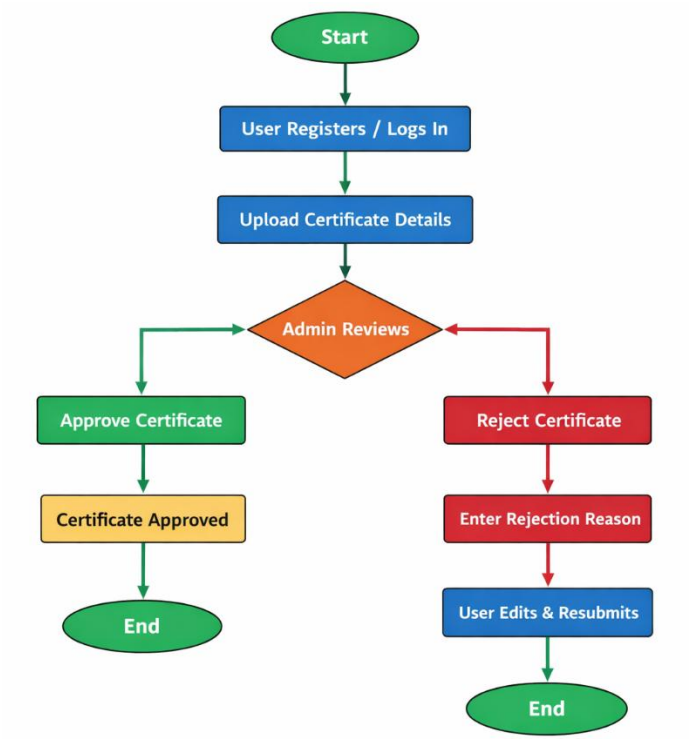


Fig. DFD Level 1 Diagram

4.4 Database Design

The system uses a relational database with three main tables:

- **Users Table:** Stores user registration details
- **Admin Table:** Stores admin login credentials
- **Certificates Table:** Stores certificate details, status, and rejection reason

Relationships:

- Each certificate is linked to a user using **user_id (foreign key)**

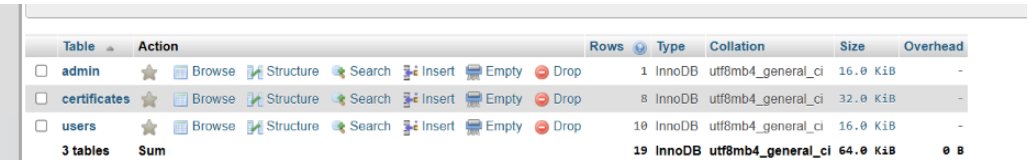


Table	Action	Rows	Type	Collation	Size	Overhead
admin	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 KiB	-
certificates	Browse Structure Search Insert Empty Drop	8	InnoDB	utf8mb4_general_ci	32.0 KiB	-
users	Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
3 tables	Sum	19	InnoDB	utf8mb4_general_ci	64.0 KiB	0 B

Fig. Database Tables

4.5 Tables Used

The FDP/STTP Digital Certificate Management System uses a relational database consisting of three main tables: **users**, **admin**, and **certificates**. Each table is designed to store specific information required for the proper functioning of the system.

1. Users Table

The **users** table stores the details of all registered users in the system.

Fields:

- **id** (Primary Key): Unique identifier for each user
- **name**: Name of the user
- **email**: User's email address (used for login)
- **password**: User's password
- **contact_no**: Contact number of the user.

Purpose:

This table is used for user registration, login authentication, and storing user profile information.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**id**	int(11)			No	*None*		AUTO_INCREMENT	Change Drop More
2	**name**	varchar(100)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
3	**email**	varchar(100)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
4	**password**	varchar(100)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More
5	**contact_no**	varchar(15)	utf8mb4_general_ci		Yes	*NULL*			Change Drop More

Table 1: -User Table

2. Admin Table

The **admin** table stores login details of the administrator.

Fields:

- **id** (Primary Key): Unique identifier for admin
- **username**: Admin username
- **password**: Admin password

Purpose:

This table is used to authenticate the admin and provide access to the admin dashboard, where certificates are managed.

Extra options

	id	username	password
Edit Copy Delete	1	admin	admin123

Check all With selected: Edit Copy Delete Export

Table 2:-Admin Table

3. Certificates Table

The **certificates** table stores all certificate-related information uploaded by users.

Fields:

- **id** (Primary Key): Unique certificate ID
- **user_id** (Foreign Key): Links certificate to a specific user
- **event_name**: Name of the FDP/STTP program
- **organizer**: Organization conducting the event.
- **academic_year**: Year of the program
- **program_type**: Type (FDP or STTP)
- **start_date**: Starting date of the program.
- **end_date**: Ending date of the program.
- **duration**: Duration in days
- **file_name**: Uploaded certificate file name.
- **status**: Status of certificate (Pending / Approved / Rejected)
- **rejection_reason**: Reason for rejection (if any)

Purpose:

This table is the core of the system. It stores all certificate details and tracks their status. It also helps the admin review and manage certificates.

	id	user_id	event_name	organizer	start_date	end_date	duration	file_name	status	academic_year	program_type	rejection_reason
Edit Copy Delete	1	3	csi	ghrcem	2026-03-06	2026-03-12	7	Screenshot 2026-03-04 233017.png	Rejected	NULL	NULL	NULL
Edit Copy Delete	2	3	hjk	ghrcem	2026-03-12	2026-03-27	16	Screenshot 2026-01-21 011359.png	Approved	NULL	NULL	NULL
Edit Copy Delete	3	4	hjk	ghrcem	2026-03-12	2026-03-27	16	Screenshot 2026-03-04 233017.png	Pending	NULL	NULL	NULL
Edit Copy Delete	4	4	hjk	ghrcem	2026-03-12	2026-03-27	15	Screenshot 2026-02-07 143724.png	Pending	2023-24	FDP	NULL
Edit Copy Delete	5	4	hjk	ghrcem	2026-03-12	2026-03-27	15	Screenshot 2026-02-07 125413.png	Pending	2024-25	FDP	NULL
Edit Copy Delete	6	6	hjk	fghj	2026-03-03	2026-03-18	16	Screenshot 2026-03-20 100122.png	Approved	2024-25	FDP	NULL
Edit Copy Delete	7	9	hjk	fghj	2026-04-01	2026-04-30	29	Screenshot 2026-03-04 233017.png	Pending	2023-24	FDP	NULL
Edit Copy Delete	8	10	ertyui	dfghjkl	2026-04-01	2026-04-30	30	Screenshot 2026-03-04 233017.png	Approved	2025-26	FDP	NULL

Table 3: - Certificates Table

5. Implementation / Working

5.1 Frontend Implementation

The frontend of the system is developed using **HTML, CSS, and JavaScript**. HTML is used to design the structure of web pages such as registration forms, login pages, dashboards, and tables. CSS is used to enhance the appearance of the system by applying styles, colours, layouts, and responsive design. JavaScript is used for client-side validation, such as checking empty fields, validating input formats, and calculating the duration between start date and end date.

The user interface includes:

- User Registration Form
- User Login Page
- Certificate Upload Form
- User Dashboard to view status.
- Admin Dashboard for managing certificates.

The frontend ensures the system is user-friendly, interactive, and easy to navigate.

5.2 Backend Implementation

The backend is implemented using PHP, which handles all server-side operations. PHP processes form data submitted by users and performed operations such as inserting data into the database, retrieving records, and updating certificate status.

Main backend functionalities include:

- User and admin authentication (login system using sessions)
- Certificate upload handling (file storage in server folder)
- Admin approval/rejection of certificates
- Updating certificate status and storing rejection reasons
- Redirecting users to appropriate pages

PHP ensures smooth communication between the frontend and the database.

5.3 Database Connectivity

The system uses MySQL for storing all data, and PHP connects to the database using MySQL functions.

- Connection is established using mysqli_connect() in db.php.
- SQL queries like INSERT, SELECT, UPDATE are used
- Data is fetched using mysqli_fetch_assoc()
- Tables used: users, admin, certificates.

The database ensures secure storage and efficient retrieval of information.

5.4 Flowchart

The flowchart represents the step-by-step process of the system starting from user login, certificate upload, admin verification, and final approval or rejection. It helps in understanding the workflow clearly.

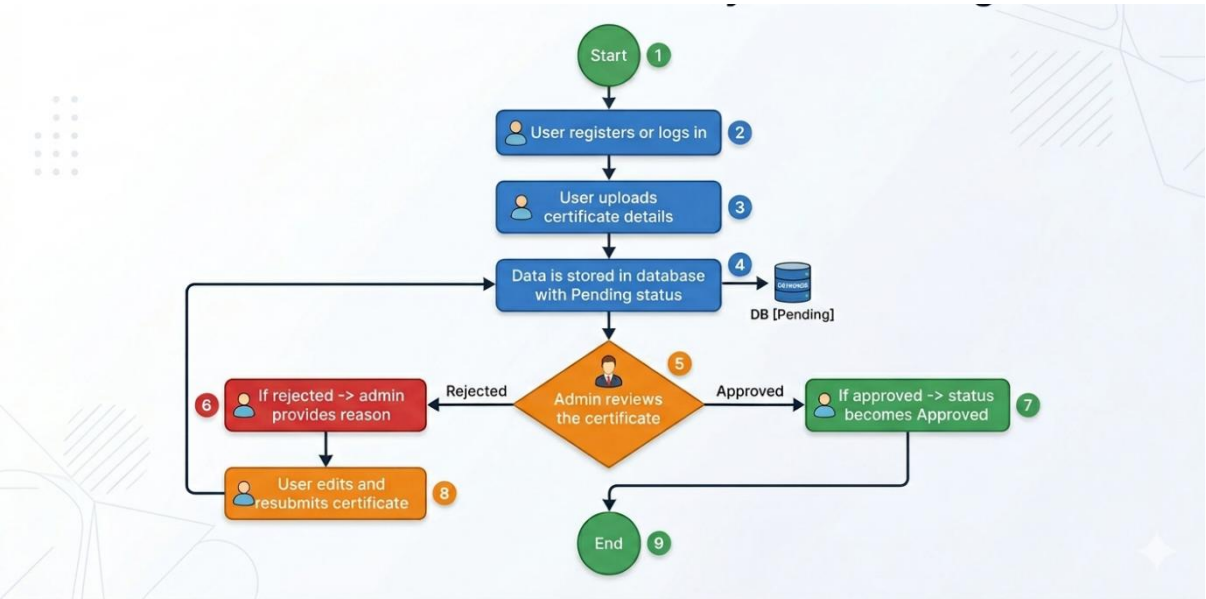


Fig. Flowchart

5.5 Working Flow

The working flow diagram explains the interaction between user and admin. It shows how certificates move from pending to approved or rejected state and how users can edit and resubmit rejected certificates.

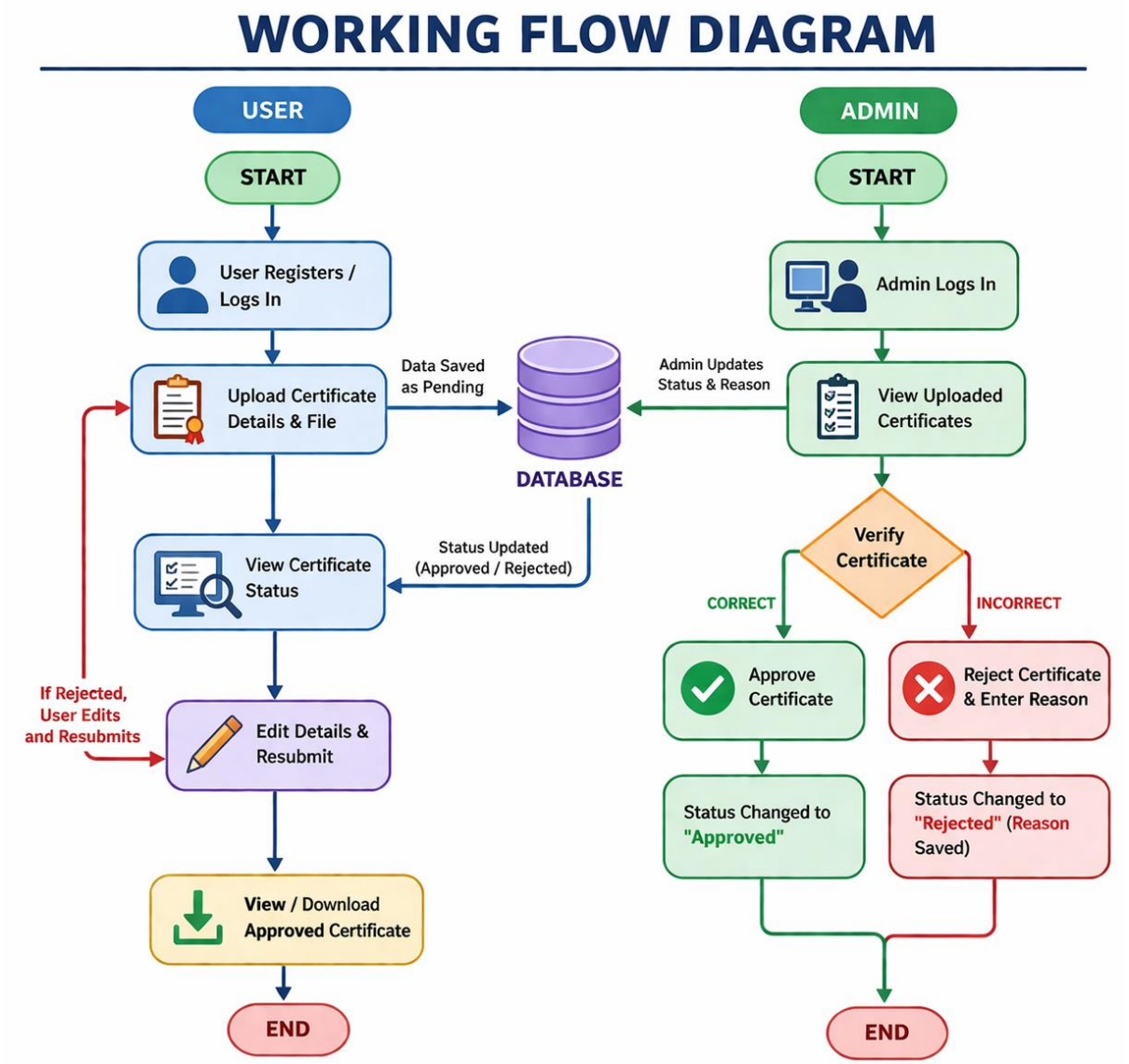
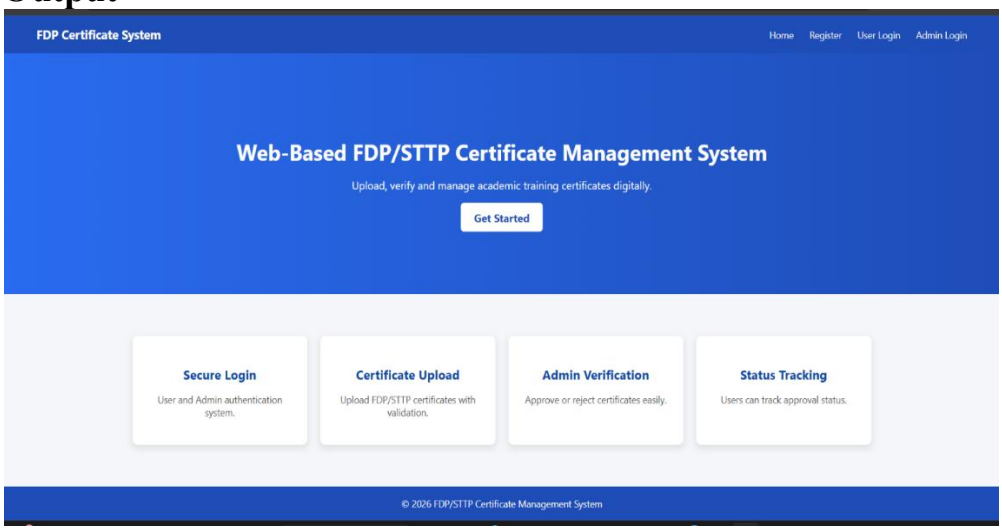


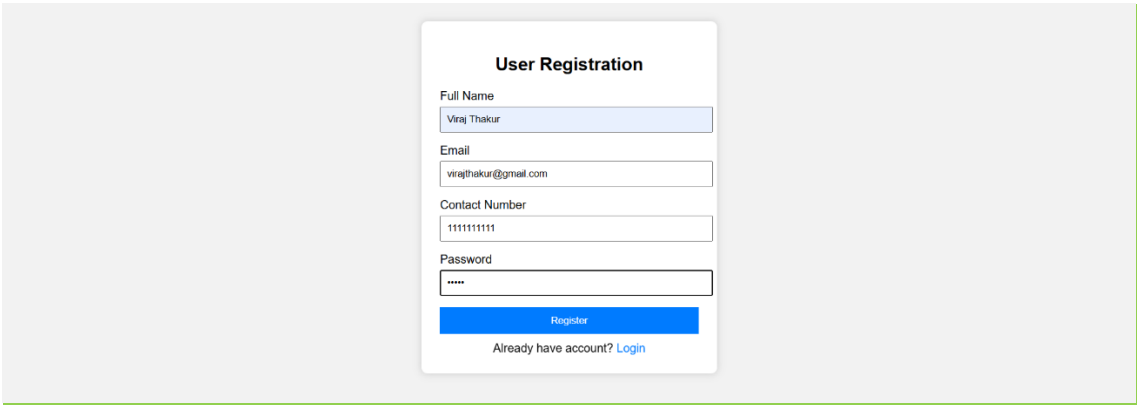
Fig 5.5: Working Flow Diagram of FDP/STTP Digital Certificate Management System

6. Results / Output

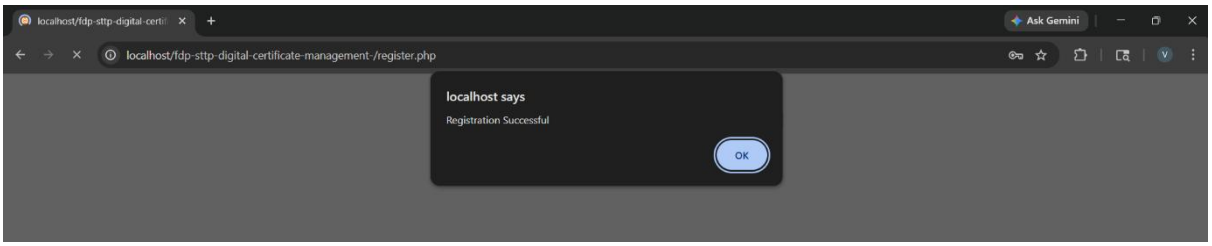
6.1 Output



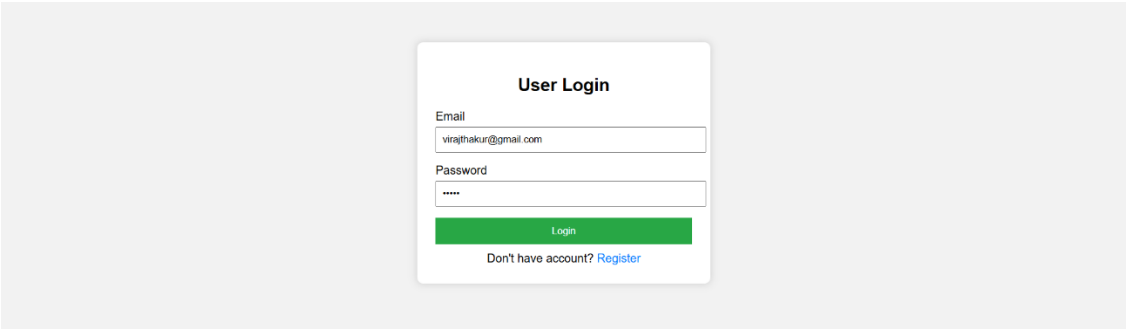
Screenshot 1: Index Page



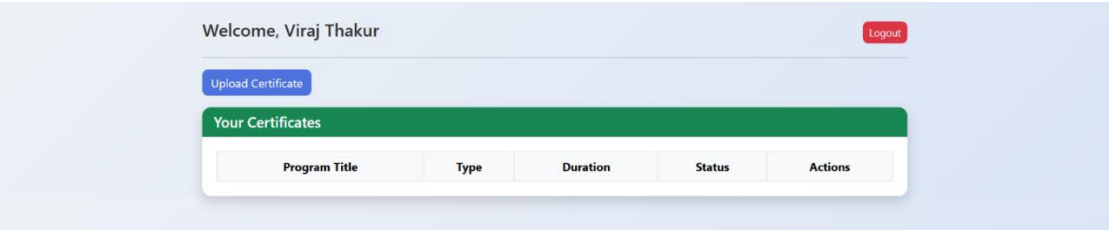
Screenshot 2: User Registration



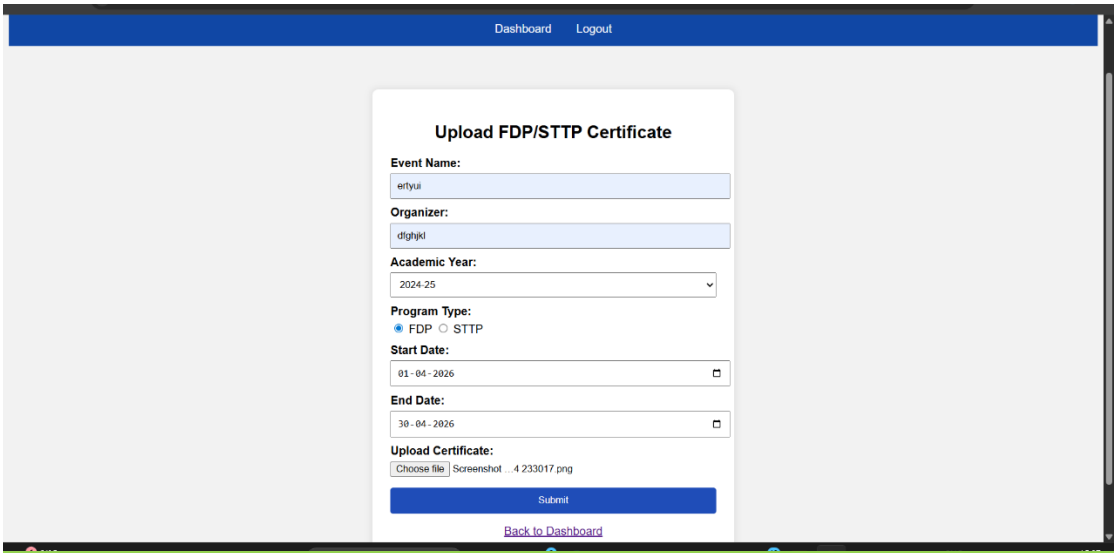
Screenshot 3: Successful registration



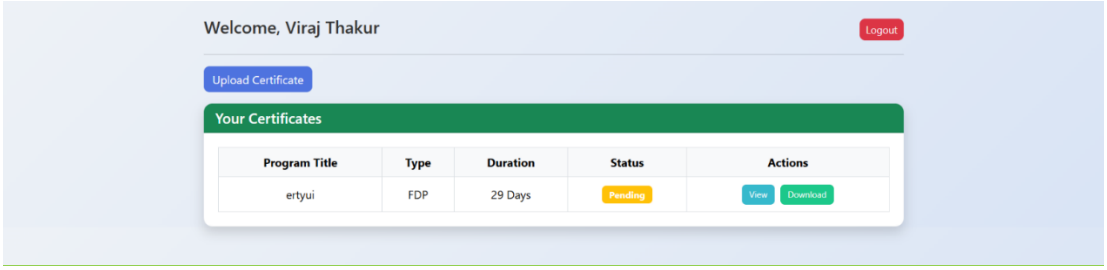
Screenshot 4: User Login



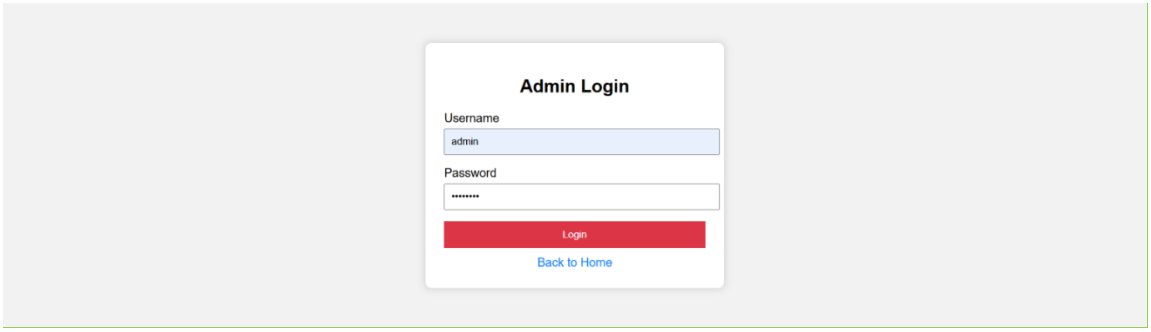
Screenshot 5: User DashBoard



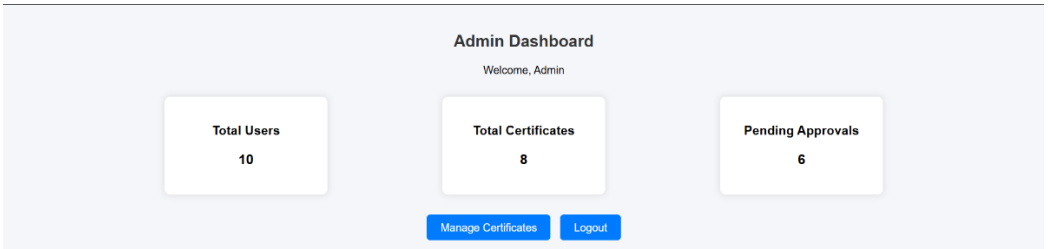
Screenshot 6: Certificate Upload Form



Screenshot 7: Show certificate in Dashboard



Screenshot 8: Admin login



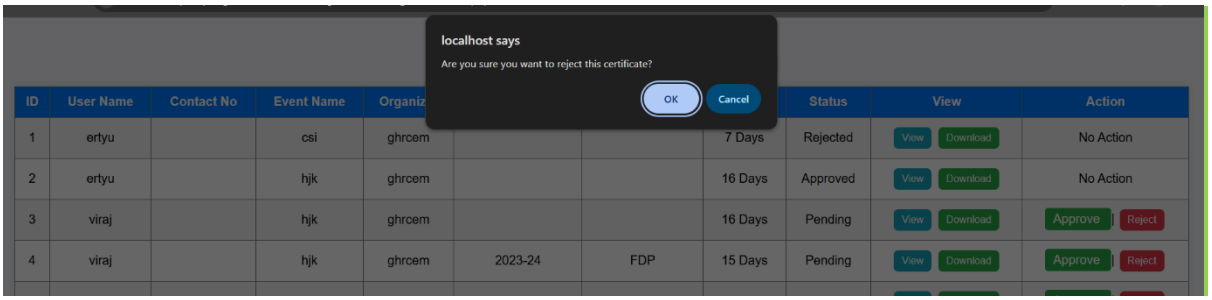
Screenshot 9: Admin Dashboard

Manage Certificates

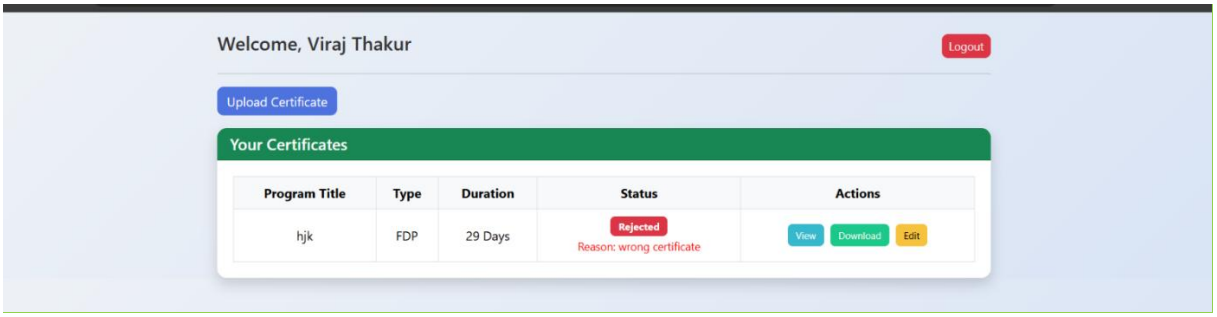
ID	User Name	Contact No	Event Name	Organizer	Academic Year	Program Type	Duration	Status	View	Action
1	ertyu		csi	ghrcem			7 Days	Rejected	View Download	No Action
2	ertyu		hjk	ghrcem			16 Days	Approved	View Download	No Action
3	viraj		hjk	ghrcem			16 Days	Pending	View Download	Approve Reject
4	viraj		hjk	ghrcem	2023-24	FDP	15 Days	Pending	View Download	Approve Reject
5	viraj		hjk	ghrcem	2024-25	FDP	15 Days	Pending	View Download	Approve Reject
6	Viraj Thakur		hjk	fghj	2024-25	FDP	16 Days	Approved	View Download	No Action
7	Viraj Thakur	1234567890	hjk	fghj	2023-24	FDP	30 Days	Rejected	View Download	No Action
8	Viraj Thakur	1111111111	ertyui	dfghjkl	2025-26	FDP	30 Days	Approved	View Download	No Action

Back to Dashboard

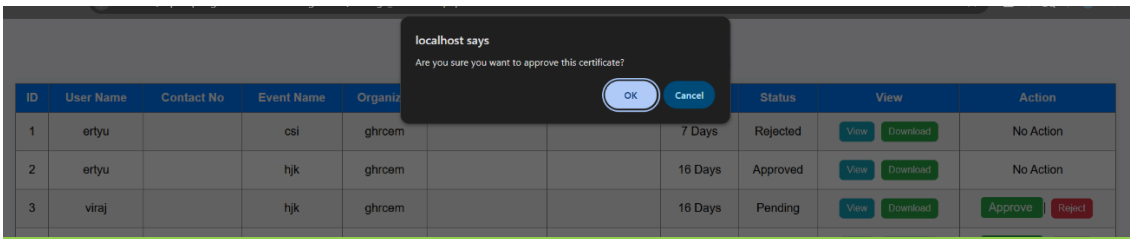
Screenshot 10: List of uploaded Certificates



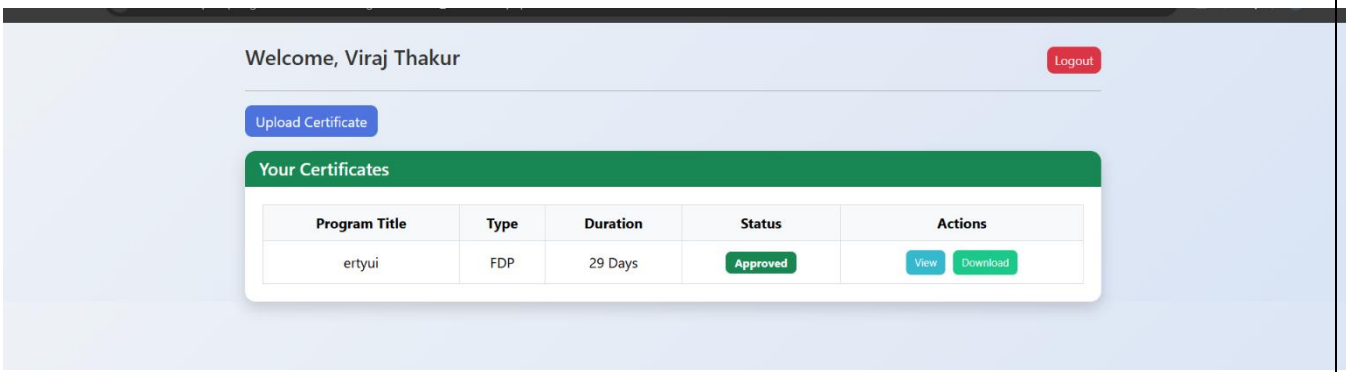
Screenshot 11: Reject Certificate



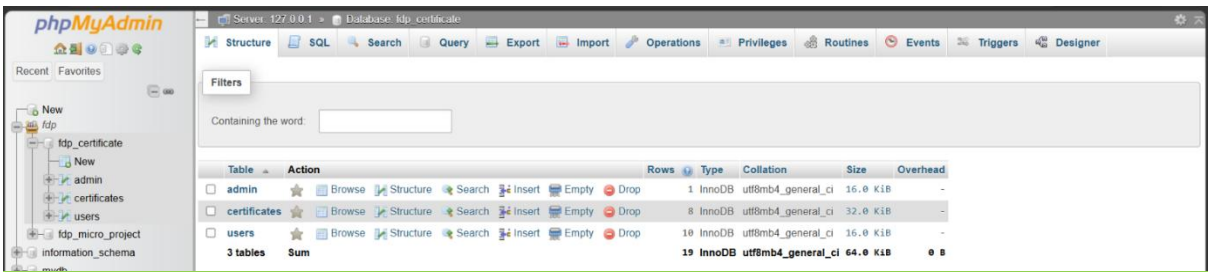
Screenshot 12: Reject status show in user Dashboard



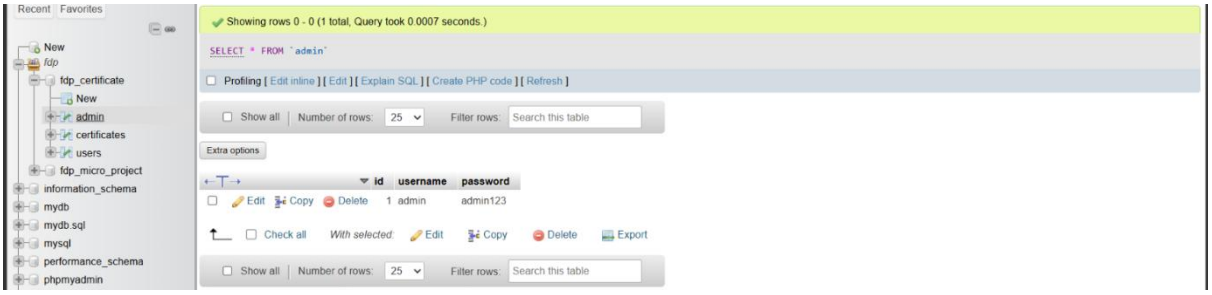
Screenshot 13: Approved certificate



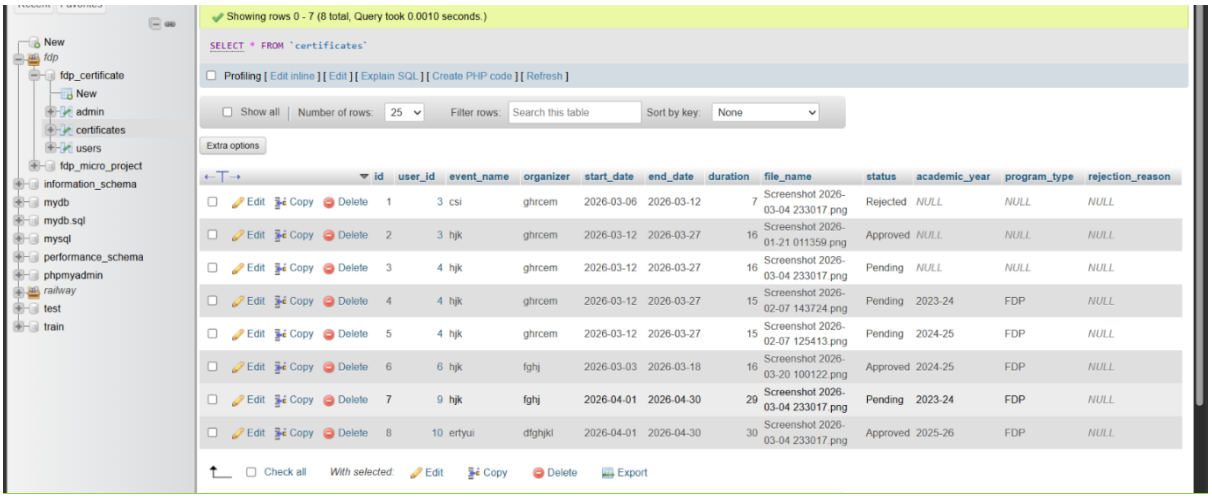
Screenshot 14: Approve status show in user Dashboard



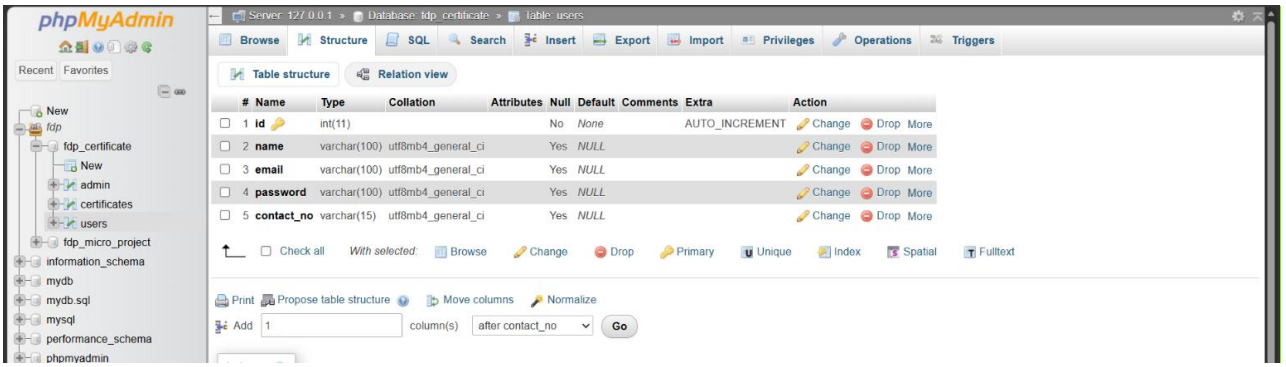
Screenshot 15: List of Table



Screenshot 16: Admin Table



Screenshot 17: Certificate Table



Screenshot 18: User table

7. Conclusion and Future Work

7.1 Conclusion

The FDP/STTP Digital Certificate Management System successfully provides a web-based solution to manage and verify certificates efficiently. The system reduces manual work by allowing users to upload certificates online and track their status. It ensures proper data storage, minimizes the risk of data loss, and improves transparency in the approval process. The admin can easily review, approve, or reject certificates with reasons, making the system organized and reliable. Overall, the project achieves its objective of simplifying certificate management and saving time.

7.2 Future Work

The system can be further enhanced by adding advanced features to improve functionality and user experience. Future improvements may include:

- Implementing **email notifications** for approval/rejection updates
- Adding **OTP-based authentication** for better security
- Integrating **QR code verification** for certificates
- Providing **mobile app support** for easier access
- Using **cloud storage** for better scalability and backup
- Adding **analytics dashboard** for admin to track reports

These enhancements will make the system more secure, scalable, and user-friendly.

8.REFERENCES

[1] PHP Manual, "PHP: Hypertext Preprocessor," The PHP Group. [Online]. Available: https://www.php.net/manual/en/. [Accessed: Apr. 2026].

[2] MySQL Documentation, "MySQL 8.0 Reference Manual," Oracle Corporation. [Online]. Available: https://dev.mysql.com/doc/refman/8.0/en/. [Accessed: Apr. 2026].

[3] MDN Web Docs, "Using the Fetch API," Mozilla Developer Network. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API/Using_Fetch. [Accessed: Apr. 2026].

[4] W3Schools, "JavaScript AJAX Introduction," W3Schools Online Web Tutorials. [Online]. Available: https://www.w3schools.com/js/js_ajax_intro.asp. [Accessed: Apr.2026].

[5] XAMPP, "XAMPP Apache + MariaDB + PHP + Perl," Apache Friends. [Online]. Available: https://www.apachefriends.org/. [Accessed: Apr. 2026].

[6] MDN Web Docs, "HTTP Cookies and Session Management," Mozilla Developer Network. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies.m [Accessed: Apr. 2026].

[7] PHP Manual, "password hash — Creates a password hash," The PHP Group. [Online]. Available:https://www.php.net/manual/en/function.password-hash.php. [Accessed: Apr. 2026].

[8] Microsoft, "Task Scheduler Overview," Microsoft Docs. [Online]. Available: https://docs.microsoft.com/en-us/windows/win32/taskschd/task-scheduler-start-page. [Accessed: Apr. 2026].

[9] W3Schools, "PHP MySQL Connect," W3Schools. [Online]. Available: https://www.w3schools.com/php/php_mysql_connect.asp. [Accessed: Apr. 2026].

[10] MDN Web Docs, "CSS Animations," Mozilla Developer Network. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Animations. [Accessed: Apr.2026].

[11] Flanagan, D., "JavaScript: The Definitive Guide," 7th ed., O'Reilly Media, 2020.

[12] Nixon, R., "Learning PHP, MySQL & JavaScript," 6th ed., O'Reilly Media, 2021.

[13] Welling, L. and Thomson, L., "PHP and MySQL Web Development," 5th ed., Addison-Wesley, 2017.

[14] Ullman, L., "PHP Advanced and Object-Oriented Programming," Peachpit Press, 2013.

[15] Gilmore, W. J., "Beginning PHP and MySQL: From Novice to Professional," 4th ed., Apress, 2010.

[16] Pratt, P. J. and Last, M. Z., "A Guide to MySQL," Cengage Learning, 2015.

[17] MDN Web Docs, "CSS Flexbox," Mozilla Developer Network. [Online]. Available: https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout/Flexbox. [Accessed: Apr. 2026].

[18] W3Schools, "PHP Sessions," W3Schools. [Online]. Available: https://www.w3schools.com/php/php_sessions.asp. [Accessed: Apr. 2026].

[19] Stack Overflow, "How to run PHP script using Windows Task Scheduler," StackOverflow Community. [Online]. Available: https://stackoverflow.com/questions/. [Accessed: Apr. 2026].

[20] GeeksforGeeks, "AJAX — Asynchronous JavaScript and XML," GeeksforGeeks. [Online]. Available: https://www.geeksforgeeks.org/ajax-introduction/. [Accessed: Apr. 2026].